

Exact Organization of Density Matrices and Entanglement Structure in the Kitaev Spin Liquid

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Motivation

- density matrix
- entanglement
-

Kitaev Model: an Introduction

- Model Hamiltonian

$$H = - \sum_{\langle ij \rangle} J_{\alpha_{ij}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} \quad ; \quad \begin{array}{c} \text{Diagram of three adjacent hexagons sharing a central vertex. The central vertex is labeled } z. \text{ The two vertices adjacent to } z \text{ are labeled } y \text{ and } x. \end{array}$$

- Majorana fermion (parton) representation: $b^x, b^y, b^z, c \Rightarrow \sigma^\mu = i b^\mu c$
- Parton Hamiltonian: define $i b_i^{\alpha_{ij}} b_j^{\alpha_{ij}} = \hat{u}_{ij}$

$$H \rightarrow \sum_{\langle ij \rangle} i J_{\alpha_{ij}} \hat{u}_{ij} c_i c_j$$

- Gutzwiller projection

$$\mathcal{P}_G = \prod_i \left(\frac{1 + D_i}{2} \right) \quad , \quad D_i = b_i^x b_i^y b_i^z c_i \Leftarrow -i \sigma_i^x \sigma_i^y \sigma_i^z$$

- Separability of the entanglement entropy (Yao and Qi, PRL **105** 080501)

$$S_A = S_A^F + S_A^G$$

Kitaev Model: Gauge Theory

In the Majorana fermion representation, the model is equivalent to a system of free Majorana fermions (matter field) coupled to a static $\mathbb{Z}_2$ gauge field

- Define the gauge field $\hat{u}_{ij} = e^{iA_{ij}}$, then the elementary $\mathbb{Z}_2$ gauge flux reads

$$e^{i\Phi} = \exp\left(i \oint_{\partial p} A\right) = \prod_{\langle ij \rangle \in \partial p} \hat{u}_{ij} = \prod_{\langle ij \rangle \in \partial p} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} =: W_p$$

- The electric field in the gauge field that flips the gauge field is $b_i^{\alpha_{ij}} \rightarrow e^{i\pi E_{ij}}$ because $b_i^{\alpha_{ij}} \hat{u}_{ij} b_i^{\alpha_{ij}} = -\hat{u}_{ij}$
- Gutzwiller projection as a Gauss law constraint: $D_i = b_i^x b_i^y b_i^z c_i = 1$. The ground state is

$$\prod_i \left(\frac{1 + D_i}{2} \right) |u\rangle |M\rangle \quad ; \quad \text{Fix the gauge: } \langle u | \hat{u}_{ij} | u \rangle = +1 \forall \langle ij \rangle$$

- Both definitions $b_i^{\alpha_{ij}} = e^{i\pi E_{ij}}$ and $b_j^{\alpha_{ij}} = e^{i\pi E_{ij}}$ are valid.
- The Gauss law is a physical constraint and cannot be broken.

Kitaev Model: Properties of Spin Correlation Functions

- Two-spin correlation functions are short-ranged (Baskaran, Mandal, and Shankar, PRL**98**, 247201)

$$\langle \sigma_i^\mu \sigma_j^\nu \rangle = \langle M | i c_i c_j | M \rangle \underbrace{\langle u | i b_i^\mu b_j^\nu | u \rangle}_{\text{vanishes if } i b_i^\mu b_j^\nu \neq \hat{u}_{ij}}$$

- Multi-spin correlation functions can ONLY be string-like

$$\left\langle \prod_{\langle ij \rangle \in \mathfrak{D}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} \right\rangle \equiv \langle \Sigma_{\mathfrak{D}} \rangle$$

- $\mathbb{Z}_2$ algebra of strings

$$\mathfrak{D} \ominus \mathfrak{D}' := \mathfrak{D} \cup \mathfrak{D}' - \mathfrak{D} \cap \mathfrak{D}' \quad : \quad \textit{illustration}$$

- String operators form a projective representation

$$\Sigma_{\mathfrak{D}} \Sigma_{\mathfrak{D}'} = \varphi(\mathfrak{D}, \mathfrak{D}') \Sigma_{\mathfrak{D} \ominus \mathfrak{D}'} \quad ; \quad \varphi(\mathfrak{D}, \mathfrak{D}') = \pm 1, \quad \varphi(\mathfrak{D}, \mathfrak{D}) = (-1)^{|\mathfrak{D}|}$$

Organization: Equivalence Relation

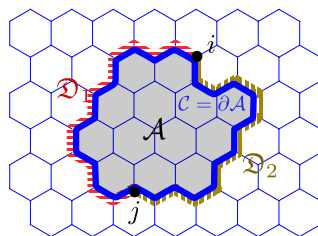
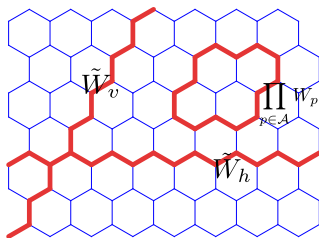
- Exact 1-form symmetry: Wilson loop $\Sigma_C |\psi_0\rangle = W_C |\psi_0\rangle = |\psi_0\rangle$
- The weights are identical up to a sign if the strings are identical up to a loop $\mathcal{C}$

$$\langle \Sigma_{\mathcal{D} \ominus \mathcal{C}} \rangle = \varphi(\mathcal{D}, \mathcal{C}) \langle \Sigma_{\mathcal{D}} \Sigma_{\mathcal{C}} \rangle = \varphi(\mathcal{D}, \mathcal{C}) \langle \Sigma_{\mathcal{D}} W_{\mathcal{C}} \rangle = \varphi(\mathcal{D}, \partial A) \langle \Sigma_{\mathcal{D}} \rangle$$

- Equivalence relation:

$$[\mathcal{D}] = \{ \mathcal{D}' : \exists \mathcal{C} \text{ composed by closed loops such that } \mathcal{D}' \ominus \mathcal{C} = \mathcal{D} \}$$

- Endpoints perspective: $\text{EP}(\mathcal{D}) = \text{EP}(\mathcal{D}')$ iff $[\mathcal{D}] = [\mathcal{D}'] \Rightarrow$ Emerging fermions at endpoints of strings



Organization of Density Matrix

- Suppose the observables are a set of $\mathbb{Z}_2$ traceless Hermitian operators: $\{\mathcal{O}_n^2 = \mathbb{1}\}$, then the density matrix can be expanded in terms of these observables

$$\rho \propto \sum_n \text{Tr}(\mathcal{O}_n^\dagger \rho) \mathcal{O}_n = \sum_n \langle \mathcal{O}_n \rangle \mathcal{O}_n$$

- For the spin-1/2 Kitaev model, the observables are Pauli strings.

$$\rho \propto \sum_{\mathfrak{D}} \langle \Sigma_{\mathfrak{D}} \rangle \Sigma_{\mathfrak{D}}$$

- Organization due to the exact 1-form Wilson symmetry:

$$\rho \propto \mathcal{P}^{(p)} \sum_{[\mathfrak{D}]} \langle \Sigma_{\mathfrak{D}} \rangle \Sigma_{\mathfrak{D}} \quad ; \quad \mathcal{P}^{(p)} := \left(\frac{\mathbb{1} + \tilde{W}_h}{2} \right) \left(\frac{\mathbb{1} + \tilde{W}_v}{2} \right) \prod_p \left(\frac{\mathbb{1} + W_p}{2} \right)$$

- Normalize $\text{Tr} \rho = 1$ to get the final result:

$$\rho = \frac{1}{2^{N-N_p}} \mathcal{P}^{(p)} \sum_{[\mathfrak{D}]} \langle \Sigma_{\mathfrak{D}} \rangle \Sigma_{\mathfrak{D}}$$

Unsolved Problems and Outlooks

- To get the REAL topological entanglement entropy, we need an exact 't Hooft symmetry and make it spontaneously broken.